

Xiaohan Cai

Department of Mathematics, Shanghai Jiao Tong University
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Research Interests

Geometric analysis, conformal geometry, positive scalar curvature

Education

Ph.D. in Mathematics

2022–Present

Shanghai Jiao Tong University

Advisor: Prof. Congming Li, Prof. Mijia Lai

- Dissertation title:
- Expected graduation: June, 2027.

B.Sc. in Mathematics

2018–2022

Shanghai Jiao Tong University

Thesis: Comparison theorems on manifolds with boundary

Supervisor: Prof. Mijia Lai

Visiting student at BICMR & Peking University

Spring 2021

Employment History

Academic Positions and Visits

Research Visitor

December 2024–May 2026

Michigan State University

Host: Prof. Xiaodong Wang

- Worked on problems related to sharp constants of Sobolev type inequalities.

Publications and Preprints

1. Xiaohan Cai, Mijia Lai: *Liouville equations on complete surfaces with nonnegative Gauss curvature*. Pacific J. Math. 332-1 (2024), 23–37. [Jounral link](#), [arXiv:2309.01956](#).
2. Xiaohan Cai, Mijia Lai: *A survey of classification results for solutions to the Liouville equation*. J. Math. Study, 57 (2024), pp. 398-424. [Jounral link](#)
3. Xiaohan Cai, Mijia Lai, Chilin Zhang: *An area growth estimate of the Liouville equation*. [arXiv:2409.19327](#).
4. Xiaohan Cai: *Uniqueness results for positive harmonic functions on manifolds with non-negative Ricci curvature and strictly convex boundary*. [arXiv: 2511.09994](#).
5. Xiaohan Cai, Xiaodong Wang: *A modified Bakry-Émery Γ_2 criterion inequality and the monotonicity of the Tsallis entropy*. [arXiv:2512.15525](#).

Selected Talks

Volume comparison from the viewpoint of isoperimetric problem

March 2023

Graduate Student Forum on Spectral Geometry

USTC, Anhui, China

Asymptotic behavior of solutions to some semilinear equations on $\mathbb{R}^2$	June 2023
<i>International Conference on PDEs and Geometric Analysis</i>	<i>SJTU, Shanghai, China</i>
Liouville equations on complete surfaces with nonnegative Gauss curvature	October 2023
<i>NJU-SJTU geometric analysis seminar</i>	<i>NJU, Nanjing, China</i>
Some classification results for solutions to Liouville equations	October 2024
<i>The 7th National PhD Student Conference on Partial Differential Equations</i>	<i>SZU, Shenzhen, China</i>
Scalar curvature integration on conformally flat manifolds with nonnegative Ricci curvature	May 2026
<i>Workshop on geometric analysis</i>	<i>SJTU, Shanghai, China</i>

Teaching Experience

Teaching Assistant, Differential Geometry	Fall 2021
<i>Shanghai Jiao Tong University</i>	
Teaching Assistant, Real Analysis	Spring 2022
<i>Shanghai Jiao Tong University</i>	

Honors and Awards

2022	Shanghai Outstanding Undergraduate Graduate
2022	Zhiyuan Outstanding Student Scholarship, Shanghai Jiao Tong University.
2025	Zhu Gongjin Scholarship, Shanghai Jiao Tong University.

Grants and Funding

Service

Skills

Languages	Mandarin Chinese (native), English.
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References

Prof. Advisor Name	Prof. Reference Name
Department of Mathematics	Department of Mathematics
Your University	Another University
advisor@example.com	reference@example.com